



VANIKA SHARMA

ENGINEERING PRODUCT DEVELOPMENT
ROBOTICS (GLOBAL MERIT SCHOLAR)

website:

<https://portfolio-tawny-ten-7fv0d2vi2z.vercel.app/>



Singapore University of Technology and Design (Class of 2028)

A fiercely motivated and results-focused individual, anchored by a solid academic foundation and a track record of thriving in demanding environments, consistently surpassing expectations. My keen analytical and problem-solving abilities are harmonized with a robust work ethic and a proactive stance toward addressing challenges. Highly organized and efficient, I excel in juggling multiple projects concurrently, prioritizing adeptly, and meeting deadlines with unwavering precision. I am enthusiastic about channeling my skills and passion into initiatives that nurture innovation, collaboration, and professional growth.

WORK EXPERIENCE

BeeX — via SGInnovate Deep Tech Talent Programme : Mechatronics Intern

August 2026 – December 2026

- Selected into a limited-intake deep-tech placement administered by SGInnovate, Singapore's government backed deep tech venture builder
- Contributed to the mechatronic design and integration of an autonomous underwater vehicle
- Detailed project scope and results are covered by a non-disclosure agreement; happy to discuss my contributions and technical approach at an NDA-appropriate level in an interview.

Lionsbot International (Griffin Labs): Robotics RnD Intern

October 2025– December 2025

- Rapid Prototyping: Refined skills in SolidWorks and Fusion 360 to design and iterate on robotic end-effectors and structural mounts.
- Startup Agility: Learned to operate within a lean, "founder-mindset" framework, prioritizing iterative debugging and fail-fast methodologies.
- Impact: Bridged the gap between laboratory simulation and real-world deployment by iterating on the physical design of the robot to accommodate complex terrain, directly influencing the final hardware architecture

RoboSub Competition : Hardware Member, Supervised by SUTD Organisation of Autonomous Robotics

Developed an Under Water Robot

- Systems Architecture: Managed the physical layout of internal electronics to optimize thermal dissipation and minimize signal interference between high-current motors and sensitive IMUs.
- Weight Optimization: Reduced overall vehicle chassis weight by 15% through generative design and strategic use of lightweight composites, allowing for increased battery payload.

Undergraduate Research Opportunity Program in SUTD

Paper on AI in Computational Education that was presented in Kohli Conference

- Data-Driven Pedagogy: Optimized automated grading scripts to identify unique student coding patterns, helping educators distinguish between collaborative learning and academic dishonesty in high-volume STEM cohorts.
- Python Algorithm Development: Engineered a custom plagiarism detection engine using Python, implementing similarity algorithms to analyze structural and semantic code similarities.

PROJECTS

Built a Digi Camera from scratch

Independently architected a bespoke digital camera, executing the complete product cycle from PCB design and embedded firmware to a custom 3D-printed enclosure.

MakanMap: Dynamic Tactile Map

Engineered a HMI (Human-Machine Interface) for the visually impaired community, utilizing a dynamic haptic array to translate complex urban layouts into accessible, real-time tactile data.

Shelfi-mus Prime(Reconfigurable shelf)

Designed and Engineered an adaptive, sensor-driven shelving system that autonomously reconfigures its internal geometry to maximize storage density for high-volume, irregular parcel overflows

SKILLS

- Design & CAD:** *Fusion 360, SolidWorks*, 3D parametric modeling, assembly design, basic FEA.
- Electronics & Embedded:** Arduino, sensors & actuators, motor control, basic, *PCB prototyping, Rapid prototyping*
- Programming & Robotics:** *C/C++, Python*, basic control systems, kinematics concepts, Arduino IDE.
- Data Annotation & AI:** Image/video labeling, annotation QA, following detailed labeling guidelines, familiarity with ML datasets.
- Engineering:** Problem solving, root-cause analysis, technical report writing, presenting design ideas, working in cross-disciplinary teams.